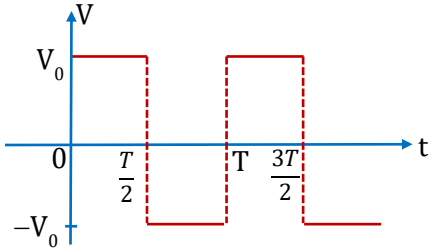
EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

A–C Circuits

1. The mean and rms value of an alternating voltage for half cycle as shown in figure are respectively



(A) V_0, V_0 (B) $\frac{V_0}{2}, V_0$ (C) $\frac{3V_0}{2}, \frac{V_0}{2}$ (D) $\frac{V_0}{4}, \frac{V_0}{2}$

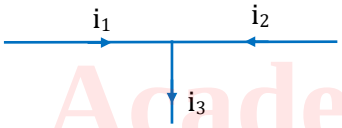
PAR051

2. In ac circuit when ac ammeter is connected it reads i current. If a student uses dc ammeter in place of ac ammeter the reading in the dc ammeter will be:

(A) $\frac{i}{\sqrt{2}}$ (B) $\sqrt{2}\,i$ (C) 0.637 i (D) zero

PAR052

3. In the given figure if $i_1 = 3 \sin \omega t$ and $i_2 = 4 \cos \omega t$, then i_3 is

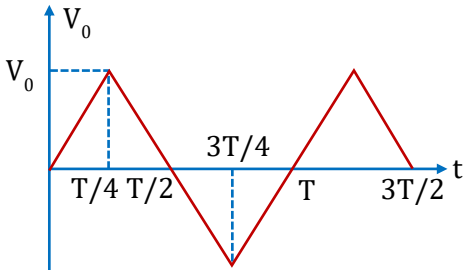


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(A) 5 sin (ωt + 53°) (B) 5 sin (ωt + 37°) (C) 5 sin (ωt + 45°) (D) 5 cos (ωt + 53°)

PAR053

4. The mean and rms value of an alternating voltage for half cycle as shown in figure is



(A) $\frac{V_0}{3}, \frac{V_0}{2}$ (B) $\frac{V_0}{\sqrt{2}}, \frac{V_0}{\sqrt{3}}$ (C) $\frac{V_0}{2}, \frac{V_0}{3}$ (D) $\frac{V_0}{2}, \frac{V_0}{\sqrt{3}}$

PAR054

5. In an ac circuit, the instantaneous voltage $e(t)$ and current $I(t)$ are given by

$$e(t) = 5[\cos \omega t + \sqrt{3} \sin \omega t] \text{ volt}, \quad i(t) = 5\left[\sin\left(\omega t + \frac{\pi}{4}\right)\right] amp$$

then:-

(A) Current leads voltage by $\frac{\pi}{4}$ (B) Voltage leads current by $\frac{\pi}{3}$

(C) Voltage leads current by $\frac{\pi}{6}$ (D) Current leads voltage by $\frac{\pi}{12}$

PAR057

For More Join: @IITJEE_Advanced

6. An alternating voltage $V = 300\sqrt{2}\sin(100t)$ is connected to a $1\mu F$ capacitor through an AC ammeter. The reading of the ammeter will be :-

(A) 10 mA (B) 30 mA (C) $30\sqrt{2}\,mA$ (D) $20\sqrt{2}\,mA$

PAR059

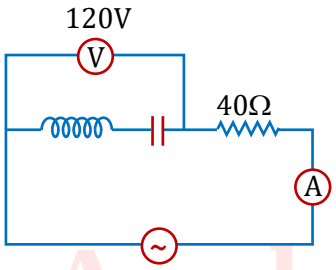
7. A series RLC circuit is driven by a generator at frequency 1000 Hz. The inductance is 90.0 mH; capacitance is 0.500 μF; and the phase constant has a magnitude of 60.0° (Take $\pi^2 = 10$) Choose **INCORRECT** statement :-

(A) Here current leads the voltage in phase (B) Here voltage leads the current in phase

(C) Resistance of circuit is $\frac{80\pi}{\sqrt{3}}\,\Omega$ (D) At resonance $\omega = \frac{\sqrt{2}}{3} \times 10^4$ rad/sec.

PAR061

8. In the given LCR series circuit find the reading of the hot wire ammeter.

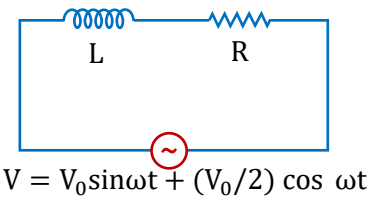


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(A) 2 A (B) 4 A (C) 5 A (D) 9 A

PAR063

9. In the given figure, the current amplitude is :-



(A) $\sqrt{\frac{5V_0^2}{4\left(\omega^2L^2+R^2\right)}}6$ (B) $\frac{V_0}{\sqrt{4\left(\omega^2L^2+R^2\right)}}$ (C) $\frac{\sqrt{5}V_0}{2\sqrt{\omega^2L^2-R^2}}$ (D) $\frac{\sqrt{5}V_0}{2\sqrt{R^2-\omega^2L^2}}$

PAR064

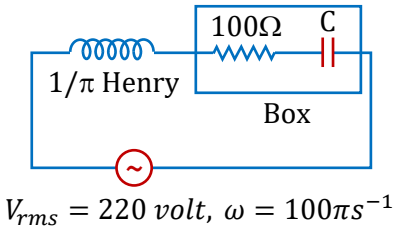
10. In series LR circuit $X_L = 3R$. Now a capacitor with $X_C = R$ is added in series. Ratio of new to old power factor is

(A) 1 (B) 2 (C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{2}$

PAR066

For More Join: @IITJEE_Advanced

11. In the circuit, as shown in the figure, if the value of R.M.S current is 2.2 ampere, the power factor of the box is



$V_{rms} = 220\ volt,\ \omega = 100\pi s^{-1}$

(A) $\frac{1}{\sqrt{2}}$ (B) 1

(C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{2}$

PAR067

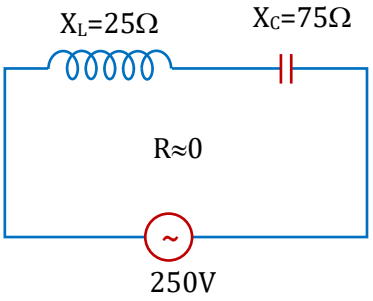
12. For shown circuit :-

(A) Current in circuit is $10A$

(B) Voltage across inductor is $100V$

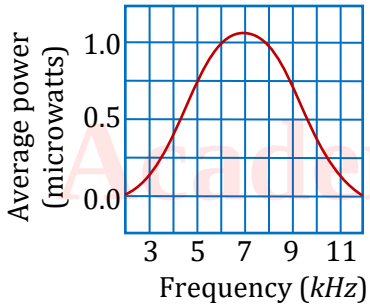
(C) Voltage across capacitor is $200V$

(D) Voltage on capacitor is more than that of supply voltage because the phase difference between V_L and V_C is 180°.



PAR070

13. The plot given below is of the average power delivered to an LRC circuit versus frequency. The quality factor of the circuit is :

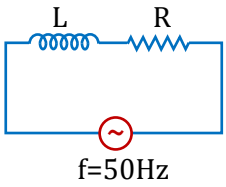


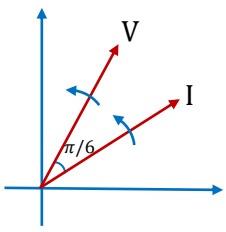
(A) 5 (B) 2.4 (C) 2.8 (D) 1.4

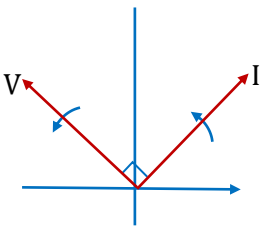
PAR074

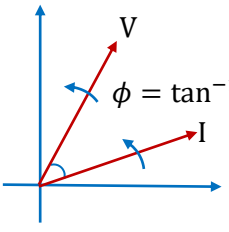
14. Consider an ideal inductor and a resistor connected to an ac source as shown. The phasor diagram of the circuit is best represented by

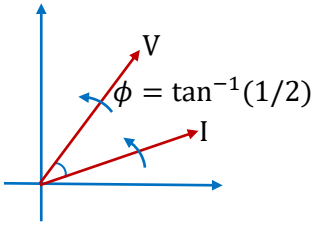
$(L = \frac{2}{\pi}mH, R = 0.15\Omega)$



(A) 

(B) 

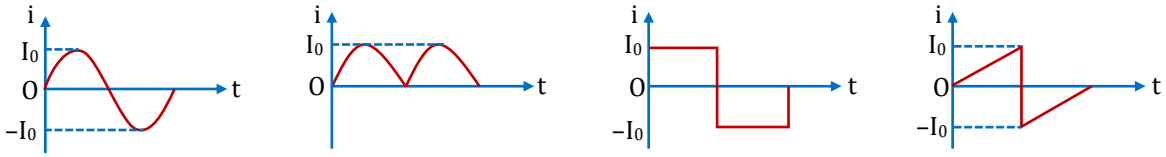
(C) 

(D) 

PAR075

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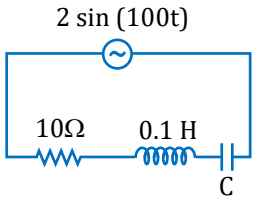
15. If I_1, I_2, I_3 and I_4 are the respective r.m.s. values of the time varying currents as shown in the four cases I, II, III and IV. Then identify the correct relations.



(A) $I_1 = I_2 = I_3 = I_4$ (B) $I_3 > I_1 = I_2 > I_4$ (C) $I_3 > I_4 > I_2 = I_1$ (D) $I_3 > I_2 > I_1 > I_4$

PAR086

16. The power factor of the circuit is $1/\sqrt{2}$. The capacitance of the circuit is equal to



(A) 400 μF (B) 300 μF (C) 500 μF (D) 200 μF

PAR087

17. When a resistance R is connected in series with an element A, the electric current is found to be lagging behind the voltage by angle θ_1. When the same resistance is connected in series with element B, current leads voltage by θ_2. When R, A, B are connected in series, the current now leads voltage by θ. Assume same AC source is used in all cases, then

(A) $\theta = \theta_2 - \theta_1$ (B) $\tan\theta = \tan\theta_2 - \tan\theta_1$

(C) $\theta = \frac{\theta_1 + \theta_2}{2}$ (D) None of these

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PAR088

18. When a voltage v_s $200\sqrt{2}$ = sin (ωt + 15°) is applied to an AC circuit the current in the circuit is found to be i = 2 sin (ω t + $\pi/4$) then average power consumed in the circuit is

(A) 200 watt (B) 400 $\sqrt{2}$ watt (C) 100 $\sqrt{6}$ watt (D) 200 $\sqrt{2}$ watt

PAR089

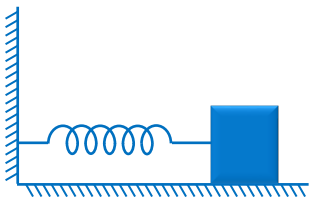
19. An ideal efficient transformer has a primary power input of 10kW. The secondary current when the transformer is on load is 25A. If the primary : secondary turns ratio is 8 : 1, then the potential difference applied to the primary coil is

(A) $\frac{10^4 \times 8^2}{25} V$ (B) $\frac{10^4 \times 8}{25} V$ (C) $\frac{10^4}{25 \times 8} V$ (D) $\frac{10^4}{25 \times 8^2} V$

PAR090

Damped and Forced Oscillations

20. In the situation shown, the block can execute free oscillation (no damping) with angular frequency ω_1. In presence of weak damping, it executes damped SHM with angular frequency ω_2. When it is subjected to a sinusoidal force, it executes forced oscillation with maximum amplitude at angular frequency ω_3 (assume damping is present) :-



(A) $\omega_1 > \omega_2 > \omega_3$ (B) $\omega_1 > \omega_2 = \omega_3$ (C) $\omega_1 = \omega_2 = \omega_3$ (D) $\omega_1 > \omega_3 > \omega_2$

PAR080

For More Join: @IITJEE_Advanced

21. A simple pendulum has a time period T if there is no air resistance. If a small air resistance is acting on the bob as it oscillates,

(A) The time period will be initially more than T and decreases with time.

(B) The time period will be less than T initially and increases with time.

(C) The time period will be less than T and remains constant.

(D) The time period will be more than T and remains constant.

PAR081

22. A pendulum with time period of $1s$ is losing energy due to damping. At certain time its energy is 45 J. If after completing 15 oscillations, its energy has become 15 J, its damping constant (in s^{-1}) is:-

(A) 2 (B) $\frac{1}{15}\ln 3$ (C) $\frac{1}{2}$ (D) $\frac{1}{30}\ln 3$

PAR083

23. A spring-mass system has undamped natural angular frequency $\omega_0 = 100\ rad\ s^{-1}$. The solution $x(t)$ at critical damping is given by $x(t) = x_0(1 + \omega_0 t)\ e^{-\omega_0 t}$, where x_0 is a constant. The system experiences the maximum damping force at time

(A) 0.01 s (B) 0.1 s (C) 0.01 πs (D) 0.1 πs

PAR084

24. An RLC circuit has a capacitance of 12 μF, an inductance of 25 mH, and a resistance of 60Ω. The current oscillates with an angular frequency of :

(A) $1.2 \times 10^3\ rad/s$ (B) $1.4 \times 10^3\ rad/s$ (C) $1.8 \times 10^3\ rad/s$ (D) $2.2 \times 10^3\ rad/s$

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PAR060

25. An RLC circuit has an inductance of $25mH$ and a capacitance of $5.0\mu F$. The charge on the capacitor does NOT oscillate but rather decays exponentially to zero. The resistance in the circuit must be:

(A) greater than or equal to $100\sqrt{2}\Omega$

(B) less than $100\sqrt{2}\Omega$ but greater than $50\sqrt{2}\Omega$

(C) less than $50\sqrt{2}\Omega$ but greater than $25\sqrt{2}\Omega$

(D) less than $25\sqrt{2}\Omega$ but greater than 0

PAR091

26. In forced oscillation of a particle the amplitude is maximum for a frequency ω_1 of the force, while the power is maximum for a frequency ω_2 of the force; then –

(A) $\omega_1 = \omega_2$

(B) $\omega_1 > \omega_2$

(C) $\omega_1 < \omega_2$ when damping is small and $\omega_1 > \omega_2$ when damping is large

(D) $\omega_1 < \omega_2$

PAR092

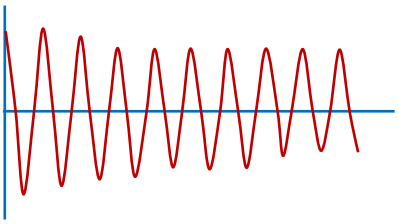
For More Join: @IITJEE_Advanced

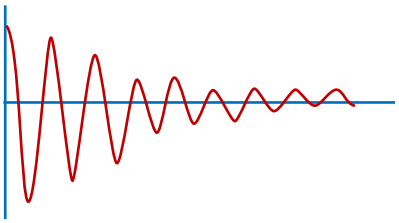
27. The angular frequency of the damped oscillator is given by $\omega = \sqrt{\frac{k}{m} - \frac{r^2}{4m^2}}$, where k is the spring constant, m is the mass of the oscillator and r is damping constant. If the ratio $\frac{r^2}{mk}$ is 8%, the change in time period composed to the undamped oscillator is approximately as follows:

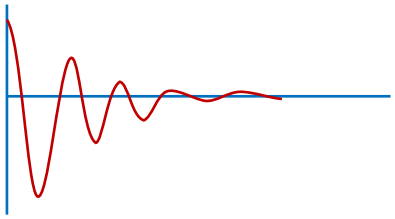
(A) decreases by 8% (B) decreases by 1% (C) increases by 1% (D) increases by 8%

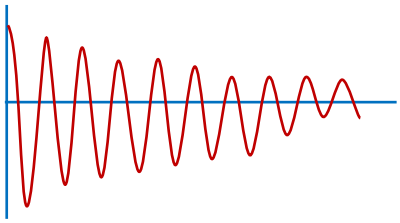
PAR093

28. Which graph has the highest Q factor ?

(A) 

(B) 

(C) 

(D) 

PAR072

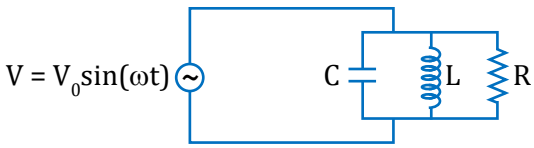
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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

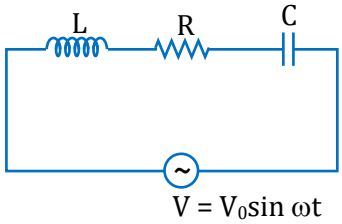
1. In the following circuit, the AC source is an ideal voltage source. Resonance is defined as a situation where the current through the source and the voltage across the source are in the same phase. At resonance, which of the following will be in the same phase as that of the applied voltage? (V_R voltage across resistor, I_R current across resistor, I_C current across capacitor, I_L current across inductor)



(A) V_R (B) I_C (C) I_L (D) I_R

PAR094

2. Consider an A-C circuit consisting of an ideal inductor (solenoid), capacitor and a resistor joined in series as shown in diagram. Choose the correct statement(s) :-



[Given : $L = 100\ mH$, $C = 10\ \mu F$, $R = 100\ \Omega$, $V = 5 \sin (500\ t)$]

(A) The circuit is in resonance for the given value of L, C, R and ω

(B) When number of turns in solenoid is doubled keeping other parameters unchanged, circuit is in resonance.

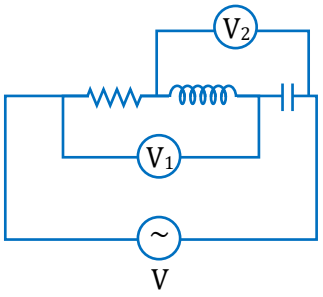
(C) When a dielectric of dielectric constant 4 is inserted in the capacitor and other parameters are kept unchanged, the circuit is in resonance.

(D) Maximum current that can flow in the circuit for given value of L,C, R and ω is 0.05 A.

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PAR095

3. In an RLC series circuit shown in fig the readings of voltmeters V_1 and V_2 are $100V$ and $120V$, respectively. The source voltage is $130V$. For this situation mark out the correct statements



(A) Voltage across resistor, inductor and capacitor are 50V, $50\sqrt{3}V$ and $120+50\sqrt{3}V$, respectively

(B) Voltage across resistor, inductor and capacitor are 50V, $50\sqrt{3}V$ and $120-50\sqrt{3}V$, respectively

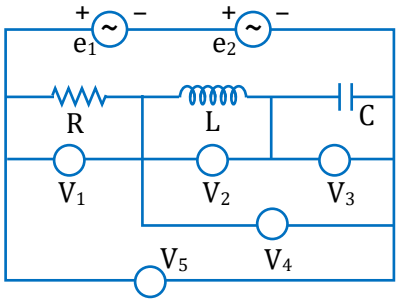
(C) Power factor of the circuit is $\frac{5}{13}$

(D) The circuit is capacitive in nature

PAR096

For More Join: @IITJEE_Advanced

4. A resistor $R = 500\ \Omega$, an ideal inductor $L = 0.9$ H and a capacitor $C = 2.5\ \mu F$ are connected with two alternating source $e_1 = 100\sin(1000t)$ and $e_2 = 100\sqrt{3}\cos(1000t)$. V_1, V_2, V_3, V_4 and V_5 are five ideal voltmeters and connected in the circuit as shown in the figure. Choose the correct option(s).



(A) Phase difference between voltages of V_3 and V_5 is $\frac{\pi}{2}+\tan^{-1}\left(\frac{9}{5}\right)$

(B) Reading of V_4 is 180 volt.

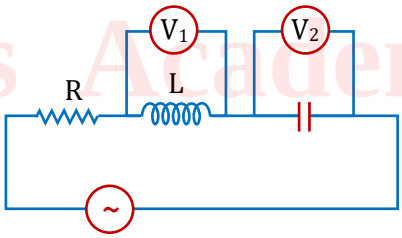
(C) Difference between reading of V_2 and V_1 is 80 volt.

(D) Ratio of reading of V_4 and V_5 is $\frac{1}{\sqrt{2}}$

PAR097

5. In the circuit shown, resistance $R = 100\,\Omega$, inductance $L=\frac{2}{\pi}\ H$ and capacitance $C=\frac{8}{\pi}\mu F$ are connected in series with an ac source of 200 volt and frequency 'f'. If the readings of the hot wire voltmeters V_1 and V_2 are same then:

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(A) $f = 125\ Hz$

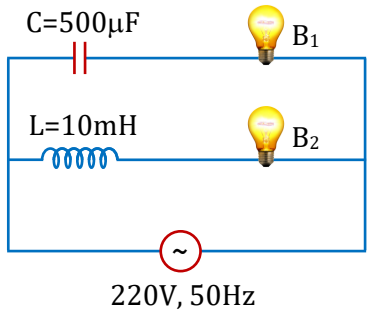
(B) $f = 250\pi\ Hz$

(C) Current through R is $2A$

(D) $V_1 = V_2 = 1000$ volt

PAR098

6. In the circuit shown in the figure, if both the bulbs B_1 and B_2 are identical :-



(A) Their brightness will be the same.

(B) B_2 will be brighter than B_1.

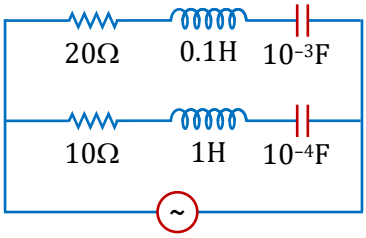
(C) As frequency of supply voltage is increased, brightness of B_1 will increase and that of B_2 will decrease.

(D) Only B_2 will glow because the capacitor has infinite impedance.

PAR125

For More Join: @IITJEE_Advanced

7. Choose the correct option(s):



$200V, \underset{(r.m.s)}{\frac{50}{\pi}} Hz$

(A) r.m.s. current drawn from the source is $30A$

(B) r.m.s. current drawn from the source is $10\sqrt{5}A$

(C) average power loss in the circuit is $6\ kW$

(D) average power loss in the circuit is zero

PAR126

COMPREHENSION TYPE QUESTIONS

PARAGRAPH FOR QUESTION NO. 8 TO 10

In a series L-R circuit, connected with a sinusoidal ac source, the maximum potential difference across L and R are respectively 3 volts and 4 volts.

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8. At an instant the potential difference across resistor is 2 volts. The potential difference in volt, across the inductor at the same instant will be :

(A) 3 cos 30° (B) 3 cos 60° (C) 3 cos 45° (D) None of these

PAR076

9. At the same instant, the magnitude of the potential difference in volt, across the ac source may be

(A) $4+3\sqrt{3}$ (B) $\frac{4+3\sqrt{3}}{2}$ (C) $1+\frac{\sqrt{3}}{2}$ (D) $2+\frac{\sqrt{3}}{2}$

PAR077

10. If the current at this instant is decreasing the magnitude of potential difference at that instant across the ac source is

(A) Increasing (B) Decreasing (C) Constant (D) Cannot be said

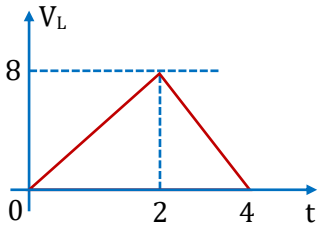
PAR078

For More Join: @IITJEE_Advanced

EXERCISE (S)

A–C Circuits

1. The potential difference across a $4H$ inductor vary with time as shown. The current is Zero at $t = 0$, then the current (A) at time $t = 2s$



PAR055

2. In an L-R series $A.C$ circuit the potential difference across an inductor and resistance joined in series are respectively $12V$ and $16V$. Find the total potential difference across the circuit.

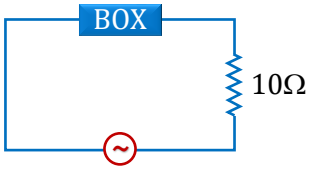
PAR056

3. An L–C–R series circuit having 220 V ac source, inductance $L = 25\ mH$ and resistance $R = 100\Omega$. If voltage across inductor is just double of voltage across resistor then the angular frequency of source is N, then find $\frac{N}{100}$.

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PAR065

4. In the circuit power factor of box is 0.5 and power factor of circuit is $\frac{\sqrt{3}}{2}$. If current leads the voltage; the value of resistance (Ω) of resistor inside the box will be :



PAR062

5. Find the value of an inductance which should be connected in series with a capacitor of $5\mu F$, a resistance of 10Ω and an ac source of 50 Hz so that the power factor of the circuit is unity.

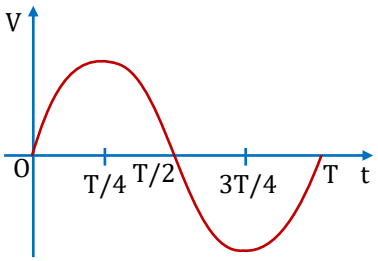
PAR069

6. A series LCR circuit containing a resistance of 120Ω has angular resonance frequency 4×10^5 rad s^{-1}. At resonance the voltages across resistance and inductance are 60 V and 40 V respectively. Find the values of L and C. At what frequency the current in the circuit lags the voltage by 45°?

PAR099

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7. In an *LR* series circuit, a sinusoidal voltage $V = V_0 \sin \omega t$ is applied. It is given that $L = 35\ mH$, $R = 11\ \Omega$, $V_{rms} = 220\ V$, $\frac{\omega}{2\pi} = 50\ Hz$ and $\pi = 22/7$. Find the amplitude of current in the steady state and obtain the phase difference between the current and the voltage. Also plot the variation of current for one cycle on the given graph.

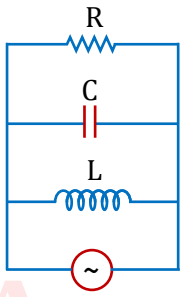


PAR100

8. A series *R-C* combination is connected to an AC voltage of angular frequency $\omega = 500$ radian/*s*. If the impedance of the *R-C* circuit is $R\sqrt{1.25}$, the time constant (in millisecond) of the circuit is :-

PAR101

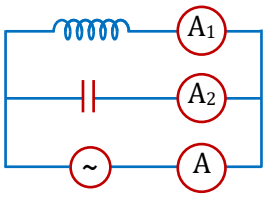
9. In the circuit shown in figure $X_L = \frac{X_C}{2} = R$, the peak value current i_0 through source is $\frac{\sqrt{x}V_0}{2R}$. Find the value of *x*.



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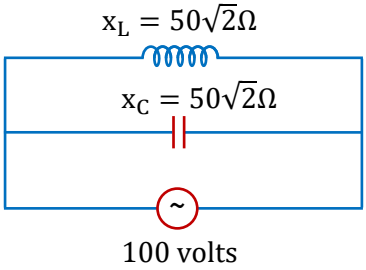
PAR127

10. Consider a circuit with an alternating source and contains inductor and capacitor. Given reading of A_1 and A_2 as 3 ampere and 5 ampere respectively. Find the reading of *A* in ampere.



PAR128

11. In an *AC* circuit capacitive reactance of capacitor is $X_C = 50\sqrt{2}\Omega$ and inductive reactance of inductor is $X_L = 50\sqrt{2}\Omega$. If the source voltage is 100 volts. Then find the current passing through the source.



PAR129

For More Join: @IITJEE_Advanced

Damped and Forced Oscillations

12. The position of a particle varies according to the law $x = A\cos(\omega t - \phi)$ under the force $F = F_0 \cos\omega\, t$. What is the average power of this force?

PAR085

13. A point performs damped oscillations with frequency $\omega = 25\ s^{-1}$. Find the damping coefficient β if at the initial moment the velocity of the point is equal to zero and its displacement from the equilibrium position is $\eta = 1.020$ times less than the amplitude at that moment.

PAR130

14. Find the quality factor of the oscillator whose displacement amplitude decreases $\eta = 2.0$ times every $n = 110$ oscillations.

PAR131

15. How soon does the current amplitude in an oscillating circuit with quality factor $Q = 5000$ decrease $\eta = 2.0$ times if the oscillation frequency is $n = 2.2\ MHz$?

PAR132

16. An oscillating circuit incorporates a leaking capacitor. Its capacitance is equal to C and active resistance to R. The coil inductance is L. The resistance of the coil and the wires is negligible. Find:

(a) the damped oscillation frequency of such a circuit;

(b) its quality factor.

PAR133

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EXERCISE - JEE (Main) PYQ

1. An arc lamp requires a direct current of $10A$ at $80V$ to function. If it is connected to a $220V$ (rms), $50Hz$ AC supply, the series inductor needed for it to work is close to: **[JEE (Main) 2016]**

(1) 0.065 H (2) 80 H (3) 0.08 H (4) 0.044 H

PAR017

2. The displacement of a damped harmonic oscillator is given by

$x(t) = e^{-0.1t}\cos(10\pi t + \phi)$. Here t is in seconds.

The time taken for its amplitude of vibration to drop to half of its initial value is close to : **[JEE (Main) 2019]**

(1) 13 s (2) 7 s (3) 27 s (4) 4 s

PAR021

3. A 750 Hz, 20 V (rms) source is connected to a resistance of 100 Ω, an inductance of 0.1803 H and a capacitance of 10 μF, all in series. The time in which the resistance (heat capacity $2J/°C$) will get heated by 10°C. (Assume no loss of heat to the surroundings) is close to : **[JEE (Main) 2020]**

(1) 418 s (2) 245 s (3) 348 s (4) 365 s

PAR022

4. A LCR circuit behaves like a damped harmonic oscillator. Comparing it with a physical spring-mass damped oscillator having damping constant 'b', the correct equivalence would be: **[JEE (Main) 2020]**

Rankers Academy JEE

(1) $L \leftrightarrow m, C \leftrightarrow \frac{1}{k}, R \leftrightarrow b$ (2) $L \leftrightarrow \frac{1}{b}, C \leftrightarrow \frac{1}{m}, R \leftrightarrow \frac{1}{k}$

(3) $L \leftrightarrow m, C \leftrightarrow k, R \leftrightarrow b$ (4) $L \leftrightarrow k, C \leftrightarrow b, R \leftrightarrow m$

PAR024

5. An inductor coil stores 64 J of magnetic field energy and dissipates energy at the rate of 640 W when a current of $8A$ is passed through it. If this coil is joined across an ideal battery, find the time constant of the circuit in seconds : **[JEE (Main) 2021]**

(1) 0.4 (2) 0.8 (3) 0.125 (4) 0.2

PAR025

6. An ac circuit has an inductor and a resistor of resistance R in series, such that $X_L = 3R$. Now, a capacitor is added in series such that $X_C = 2R$. The ratio of new power factor with the old power factor of the circuit is $\sqrt{5}:x$. The value of x is ____. **[JEE (Main) 2021]**

PAR026

7. An AC current is given by $I = I_1 \sin\omega t + I_2 \cos\omega t$. A hot wire ammeter will give a reading : **[JEE (Main) 2021]**

(1) $\sqrt{\frac{I_1^2 - I_2^2}{2}}$ (2) $\sqrt{\frac{I_1^2 + I_2^2}{2}}$ (3) $\frac{I_1 + I_2}{\sqrt{2}}$ (4) $\frac{I_1 + I_2}{2\sqrt{2}}$

PAR134

For More Join: @IITJEE_Advanced

8. Match List-I with List-II:

	List-I		List-II
(a)	Phase difference between current and voltage in a purely resistive AC circuit	(i)	$\frac{\pi}{2}$; current leads voltage
(b)	Phase difference between current and voltage in a pure inductive AC circuit	(ii)	Zero
(c)	Phase difference between current and voltage in a pure capacitive AC circuit	(iii)	$\frac{\pi}{2}$; current lags voltage
(d)	Phase difference between current and voltage in an LCR series circuit	(iv)	$\tan^{-1}\left(\frac{X_C - X_L}{R}\right)$

Choose the most appropriate answer from the options given below : **[JEE (Main) 2021]**

(1) (a)–(i),(b)–(iii),(c)–(iv),(d)–(ii) (2) (a)–(ii),(b)–(iv),(c)–(iii),(d)–(i)

(3) (a)–(ii),(b)–(iii),(c)–(iv),(d)–(i) (4) (a)–(ii),(b)–(iii),(c)–(i),(d)–(iv)

PAR135

9. Given below are two statements :

Statement-I : The reactance of an ac circuit is zero. It is possible that the circuit contains a capacitor and an inductor.

Statement-II : In ac circuit, the average power delivered by the source never becomes zero.

In the light of the above statements, choose the correct answer from the options given below :

[JEE (Main) 2022]

Rankers Academy JEE

(1) Both Statement I and Statement II are true. (2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II in false. (4) Statement I is false but Statement II is true.

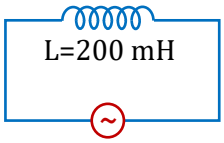
PAR027

10. A resistance of 40 Ω is connected to a source of alternating current rated 220 *V*, 50 *Hz*. Find the time taken by the current to change from its maximum value to *rms* value : **[JEE (Main) 2022]**

(1) 2.5 *ms* (2) 1.25 *ms* (3) 2.5 *s* (4) 0.25 *s*

PAR028

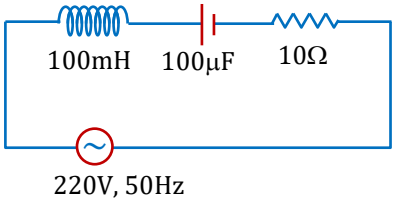
11. As shown in the figure an inductor of inductance 200 *mH* is connected to an *AC* source of emf 220 *V* and frequency 50 *Hz*. The instantaneous voltage of the source is 0 *V* when the peak value of current is $\frac{\sqrt{a}}{\pi}A$. The value of a is _____.



[JEE (Main) 2022]

PAR029

12. In a series LCR circuit, the inductance, capacitance and resistance are $L = 100mH$, $C = 100\mu F$ and $R = 10\Omega$ respectively. They are connected to an AC source of voltage $220V$ and frequency of 50 *Hz*. The approximate value of current in the circuit will be_____ *A*.

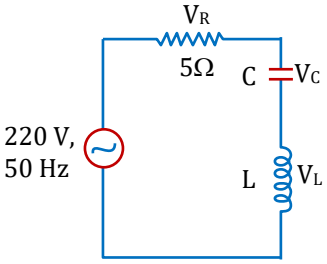


[JEE (Main) 2022]

PAR136

For More Join: @IITJEE_Advanced

13. In the given circuit, the magnitude of V_L and V_C are twice that of V_R. Given that $f = 50\ Hz$, the inductance of the coil is $\frac{1}{K\pi} mH$. The value of K is _____. **[JEE (Main) 2022]**



PAR137

14. A series LCR circuit is connected to an ac source of 220V, $50Hz$. The circuit contain a resistance $R = 100\Omega$ and an inductor of inductive reactance $X_L = 79.6\ \Omega$. The capacitance of the capacitor needed to maximize the average rate at which energy is supplied will be _____ μF.

[JEE (Main) 2023]

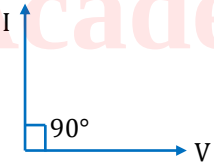
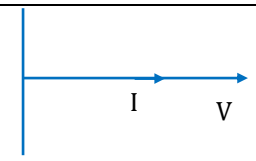
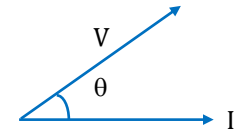
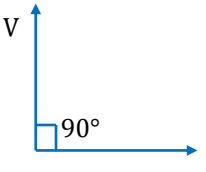
PAR103

15. A series combination of resistor of resistance 100Ω, inductor of inductance $1H$ and capacitor of capacitance $6.25\mu F$ is connected to an ac source. The quality factor of the circuit will be _____.

[JEE (Main) 2023]

PAR104

16. Match List I with List II

	List-I		**List-II**
A.	Purely capacitive circuit	I.	
B.	Purely inductive circuit	II.	
C.	LCR series at resonance	III.	
D.	LCR series circuit	IV.	

Choose the correct answer from the options given below : **[JEE (Main) 2024]**

(1) A-I, B-IV, C-III, D-II
(2) A-IV, B-I, C-III, D-II
(3) A-IV, B-I, C-II, D-III
(4) A-I, B-IV, C-II, D-III

PAR138

For More Join: @IITJEE_Advanced

17. When a coil is connected across a 20 V dc supply, it draws a current of 5 A. When it is connected across 20 V, 50 Hz ac supply, it draws a current of 4 A. The self inductance of the coil is _____ mH. (Take $\pi = 3$) **[JEE (Main) 2024]**

PAR139

18. Given below are two statements :

Statement I : In an LCR series circuit, current is maximum at resonance.

Statement II : Current in a purely resistive circuit can never be less than that in a series LCR circuit when connected to same voltage source.

In the light of the above statements, choose the ***correct*** from the options given below : **[JEE (Main) 2024]**

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

PAR141

19. In a series LCR circuit, a resistor of 300 Ω, a capacitor of 25 nF and an inductor of 100 mH are used. For maximum current in the circuit, the angular frequency of the ac source is ____ × 10^4 radians s^{-1}. **[JEE (Main) 2025]**

PAR163

20. An alternating current is given by

$$I = I_A \sin \omega t + I_B \cos \omega t$$

The r.m.s. current will be :- **[JEE (Main) 2025]**

(1) $\sqrt{I_A^2 + I_B^2}$ (2) $\frac{\sqrt{I_A^2 + I_B^2}}{2}$ (3) $\sqrt{\frac{I_A^2 + I_B^2}{2}}$ (4) $\frac{|I_A + I_B|}{\sqrt{2}}$

PAR164

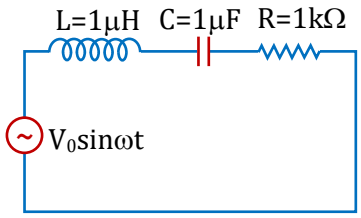
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EXERCISE - JEE (Advanced) PYQ

1. In the circuit shown, $L = 1\ \mu H$, $C = 1\ \mu F$ and $R = 1\ k\Omega$. They are connected in series with an *A.C.* source $V=V_0 \sin \omega t$ as shown. Which of the following options is/are correct ?

[JEE (Advanced) 2017]



(A) The frequency at which the current will be in phase with the voltage is independent of R.

(B) At $\omega \sim 0$ the current flowing through the circuit becomes nearly zero.

(C) At $\omega >> 10^6$ *rad-s*$^{-1}$, the circuit behaves like a capacitor.

(D) The current will be in phase with the voltage if $\omega = 10^4$ *rad-s*$^{-1}$.

PAR042

2. The instantaneous voltages at three terminals marked X, Y and Z are given by

$$V_X = V_0 \sin \omega t$$

$$V_Y = V_0 \sin\left(\omega t + \frac{2\pi}{3}\right) \text{ and}$$

$$V_Z = V_0 \sin\left(\omega t + \frac{4\pi}{3}\right)$$

An ideal voltmeter is configured to read *rms* value of the potential difference between its terminals. It is connected between points X and Y and then between Y and Z. The reading(s) of the voltmeter will be:- **[JEE (Advanced) 2017]**

(A) $V_{XY}^{rms} = V_0$

(B) $V_{YZ}^{rms} = V_0\sqrt{\frac{1}{2}}$

(C) Independent of the choice of the two terminals

(D) $V_{XY}^{rms} = V_0\sqrt{\frac{3}{2}}$

PAR043

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For More Join: @IITJEE_Advanced

QUESTION STEM FOR QUESTION NOS. 3 AND 4

In a circuit, a metal filament lamp is connected in series with a capacitor of capacitance $C\ \mu F$ across a $200V$, $50Hz$ supply. The power consumed by the lamp is $500W$ while the voltage drop across it is $100V$. Assume that there is no inductive load in the circuit. Take *rms* values of the voltages. The magnitude of the phase-angle (in degrees) between the current and the supply voltage is j. Assume, $\pi\sqrt{3} \approx 5$. **[JEE (Advanced) 2021]**

3. The value of C is ___.

PAR048

4. The value of j is ___.

PAR049

5. A series LCR circuit is connected to a 45 sin (ωt) Volt source. The resonant angular frequency of the circuit is 10^5 rad s^{-1} and current amplitude at resonance is I_0. When the angular frequency of the source is $\omega = 8 \times 10^4$ rad s^{-1}, the current amplitude in the circuit is $0.05\ I_0$. If $L = 50\ mH$, match each entry in List-I with an appropriate value from List-II and choose the correct option.

	List-I		**List-II**
(P)	I_0 in mA	(1)	44.4
(Q)	The quality factor of the circuit	(2)	18
(R)	The bandwidth of the circuit in rad s^{-1}	(3)	400
(S)	The peak power dissipated at resonance in Watt	(4)	2250
		(5)	500

[JEE (Advanced) 2023]

(A) $P \to 2, Q \to 3, R \to 5, S \to 1$

(B) $P \to 3, Q \to 1, R \to 4, S \to 2$

(C) $P \to 4, Q \to 5, R \to 3, S \to 1$

(D) $P \to 4, Q \to 2, R \to 1, S \to 5$

PAR105

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	D	A	D	D	B	A	B	A	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	D	D	C	B	C	B	C	B	A
Que.	21	22	23	24	25	26	27	28		
Ans.	D	D	A	B	A	D	C	A		

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	AD	BC	ACD	CD	ACD	BC	AC	A	B	A

EXERCISE (S)

1. 2 **2.** $20V$ **3.** 80

4. 5 **5.** $\frac{20}{\pi^2} \cong 2H$ **6.** $0.2\ mH,\ \frac{1}{32}\ \mu F, 8 \times 10^5\ \text{rad}/s$

7. $20\ A, \pi/4,$ **8.** 4 **9.** 5

10. 2 **11.** 0 **12.** $\frac{F_0 A\omega}{2}\sin\phi$

13. $\beta = \omega\sqrt{\eta^2 - 1} = 5s^{-1}$ **14.** $Q = \pi n/\ln\eta = 5 \times 10^2$ **15.** $t = \frac{Q}{\pi v}\ln\eta = 0.5ms$

16. (a) $\omega = \sqrt{\frac{1}{LC} - \frac{1}{4R^2C^2}}$ (b) $Q = \frac{1}{2}\sqrt{\frac{4R^2C}{L} - 1}$

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	2	3	1	4	1	2	4	3	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	242	22	0	40	4	4	10	3	2	3

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5
Ans.	AB	CD	100.00	60.00	B

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